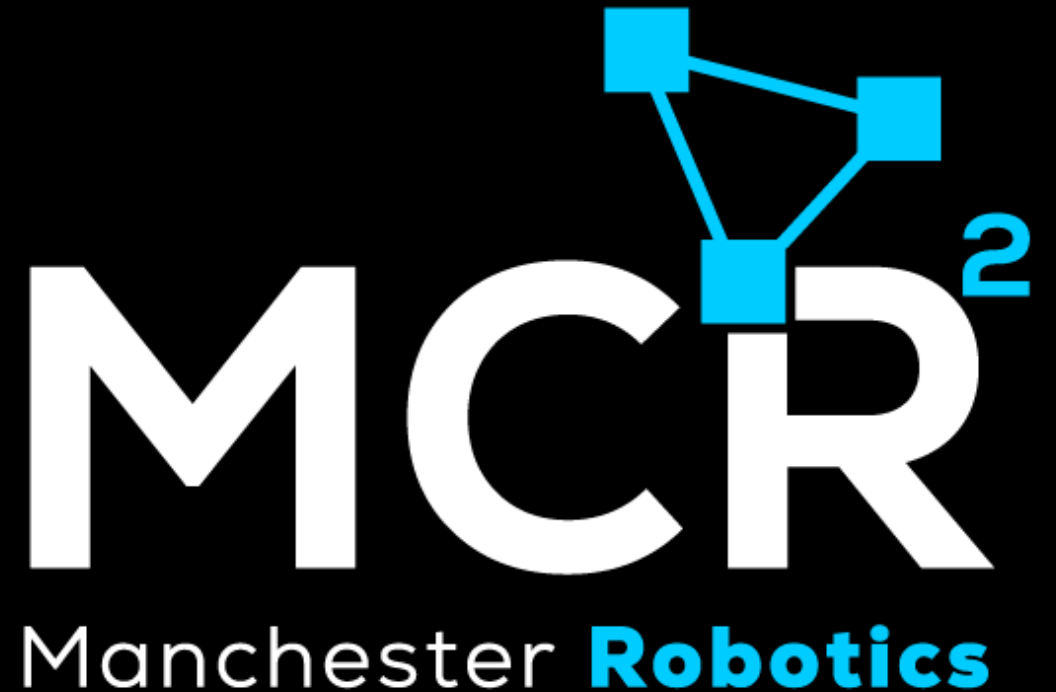


{Learn, Create, Innovate};

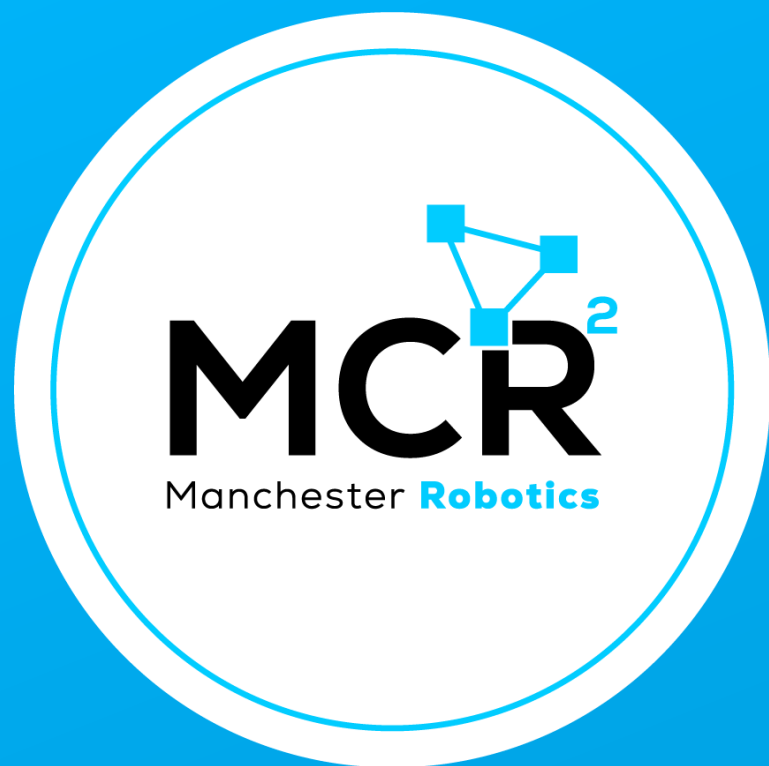
Democratising Robotics Education

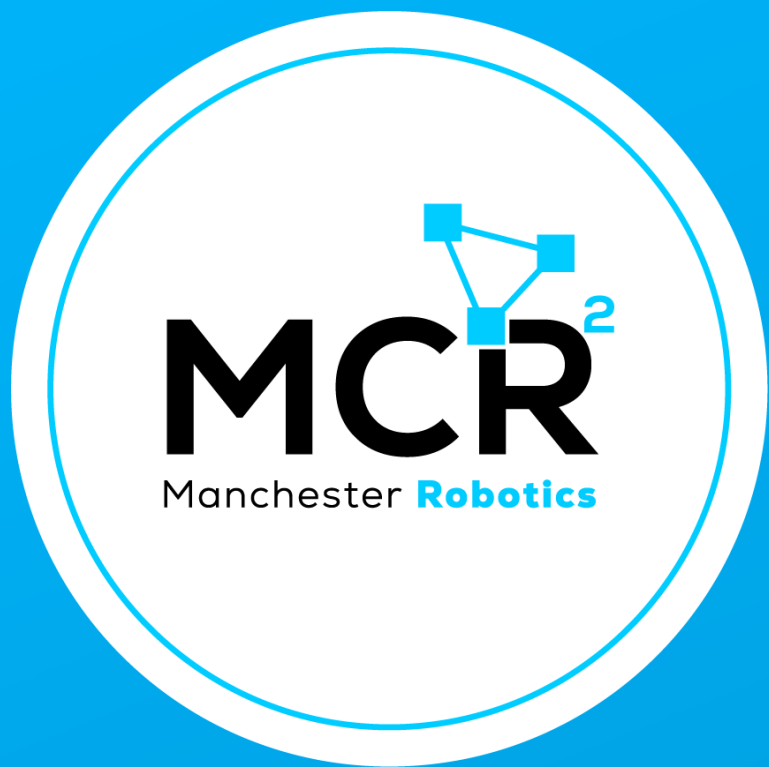
Dr. Mario Martinez

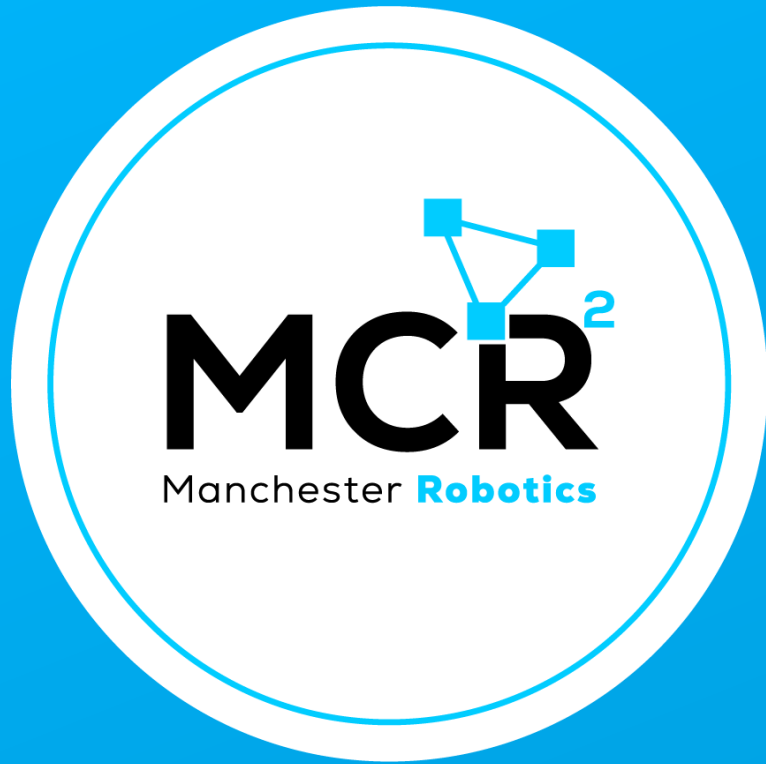
CTO & Co-Founder



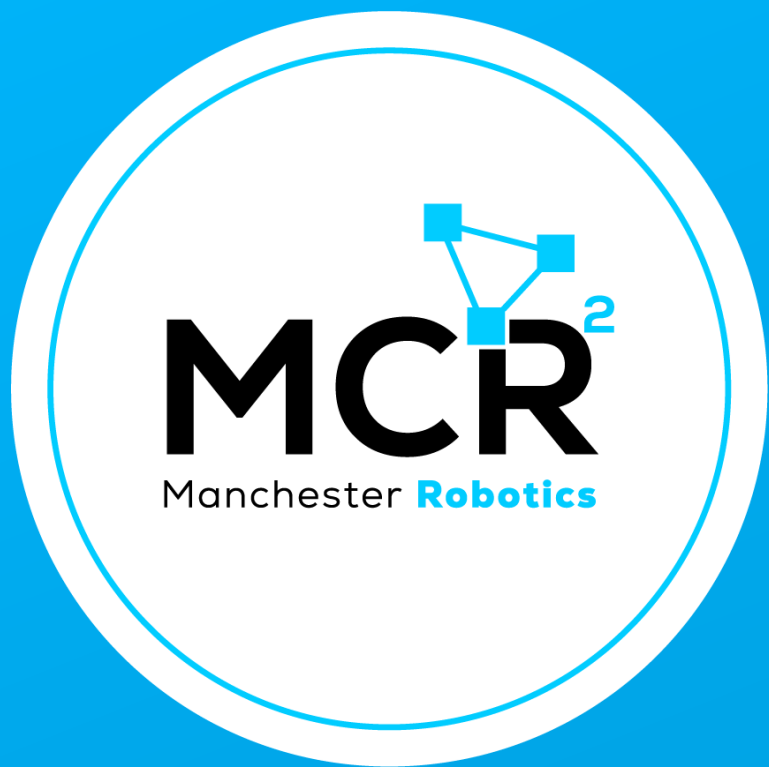


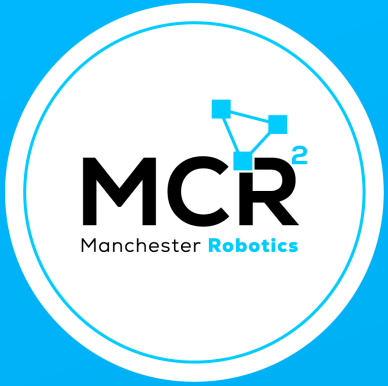


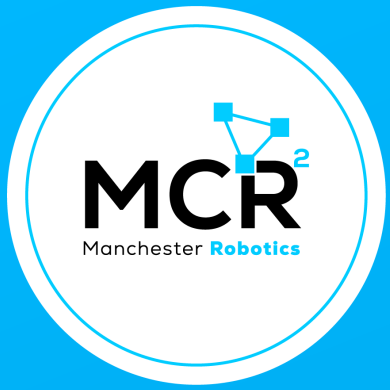




Robotics For Everyone







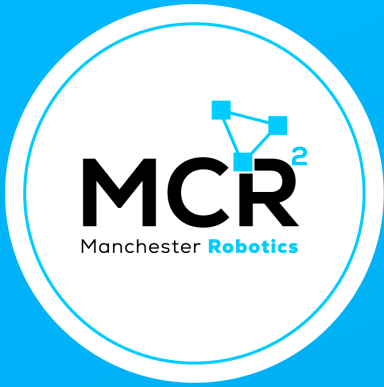


Who We Are

Manchester Robotics Limited was born as a spinout company from The University of Manchester in the United Kingdom.

It was an initiative of the Robotics Research Group within the Aerospace Research Institute.





The Team



**Gillian
Kavanagh**

Innovate UK Edge
IGS



**Dr. Alexandru
Stancu**

CEO, Co-Founder



**Prof.
Constantinos
Soutis**

Director, Co-Founder



**Dr. Mario
Martinez**

CTO, Co-Founder



**Dr. Eduard
Codres**

Head of Innovation
and Research, Co-
Founder



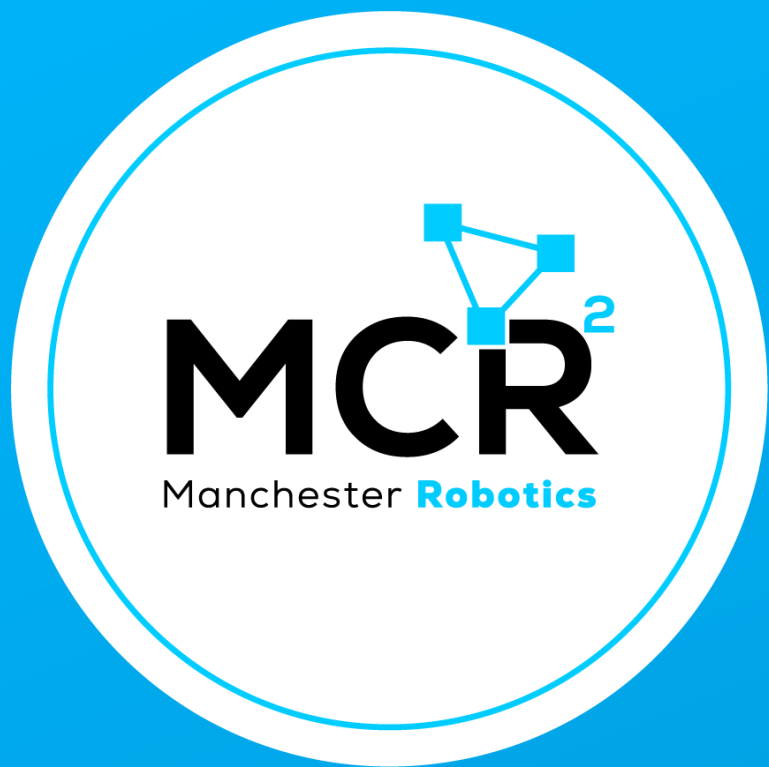
Andrew Taylor

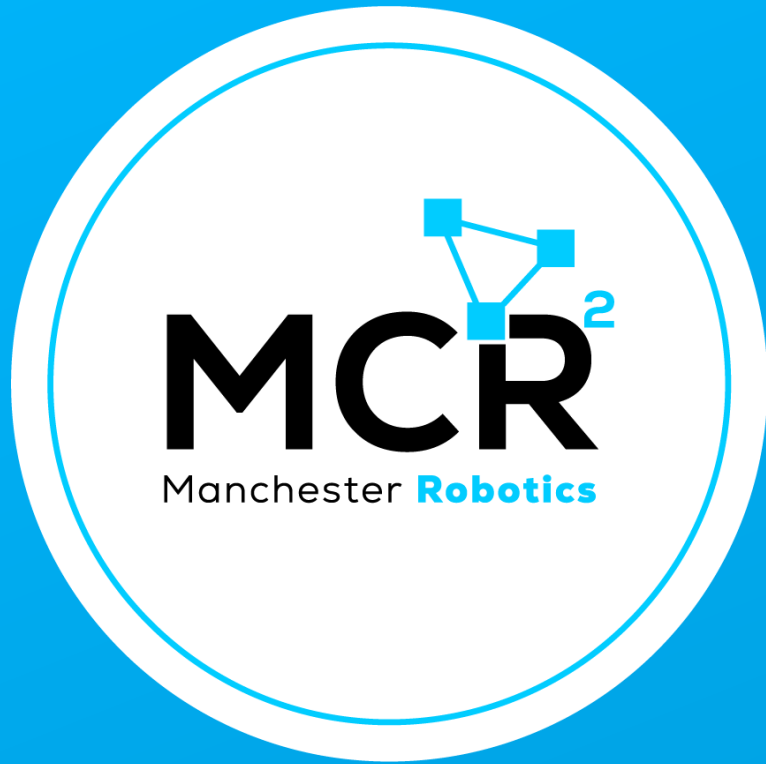
Commercial Lead



Phil Kemp

Business Advisor





Our Story



Our Story

- Educational Robotic platforms were not affordable.
- Robotic education started to become inaccessible for students and universities.
- So, as researchers in robotics, we built our own: Affordable, reliable, and just as capable.
- We didn't stop at hardware—we provide a full robotics curriculum that made NVIDIA interested in partnering with us.

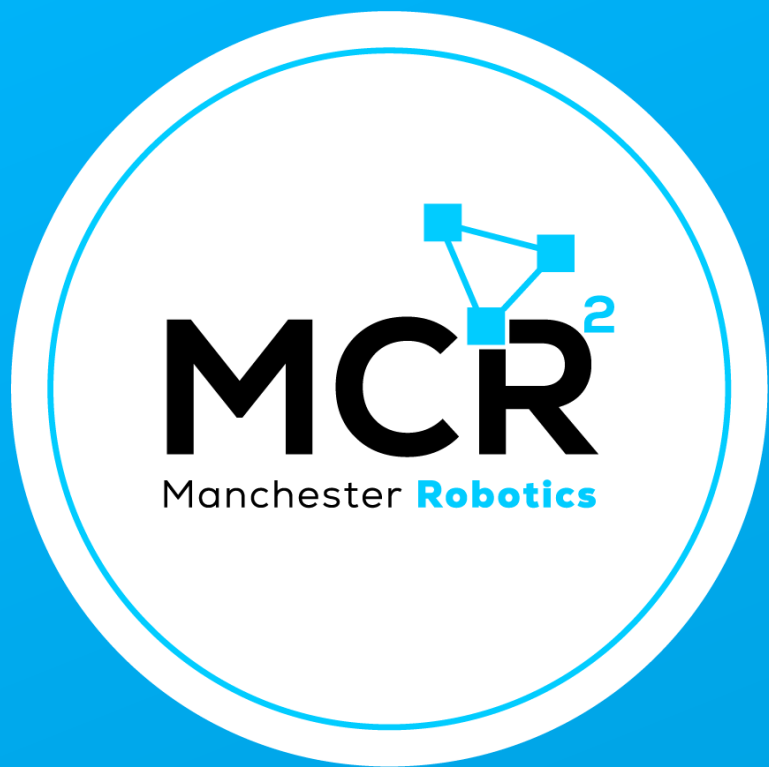
What Does It Take to Become
a **World-Class Robotics
Engineer?**

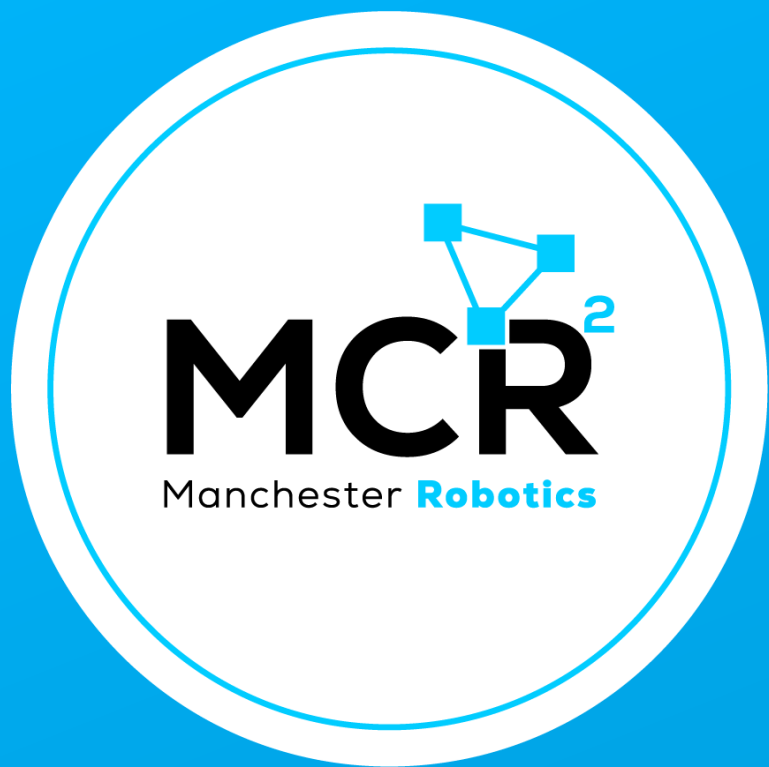
“

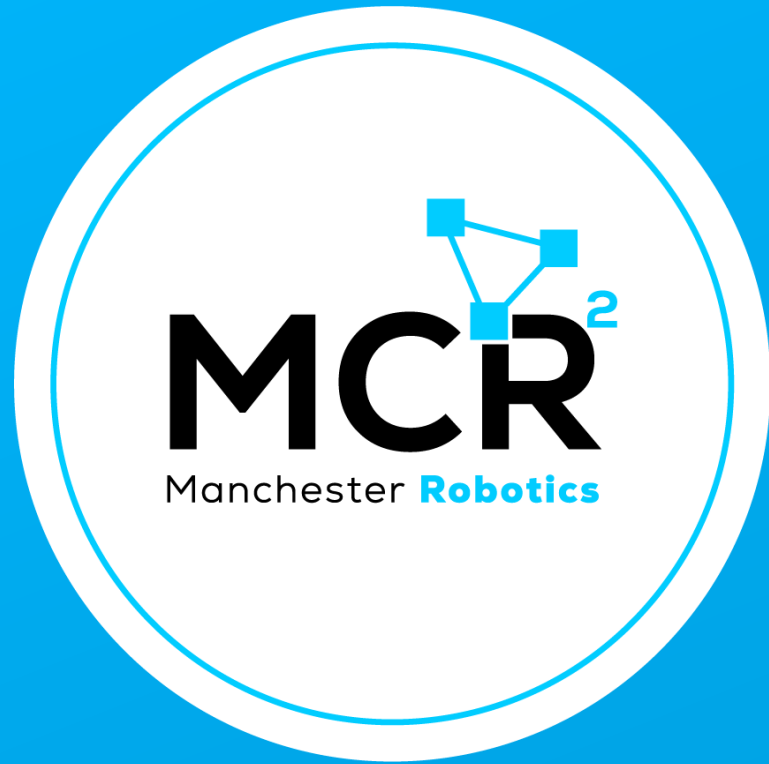
“For us robotic democratisation is not a concept... its our way of thinking, working... being.”

Professor Constantinos Soutis, Director

”







Problems We Solve



Problems We Solve

“Education is evolving so we need to evolve with it ...”

Dr Alexandru Stancu, CEO/Director

Universities

- Fragmented Teaching
- No problem Solving
- Lack of robotic platforms
- The bigger picture is lost

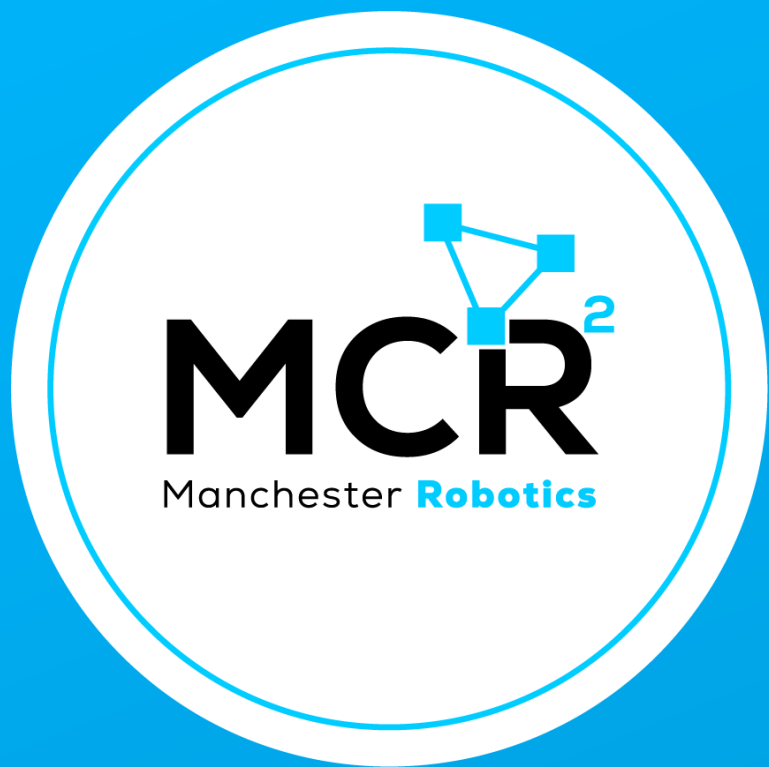
Industries

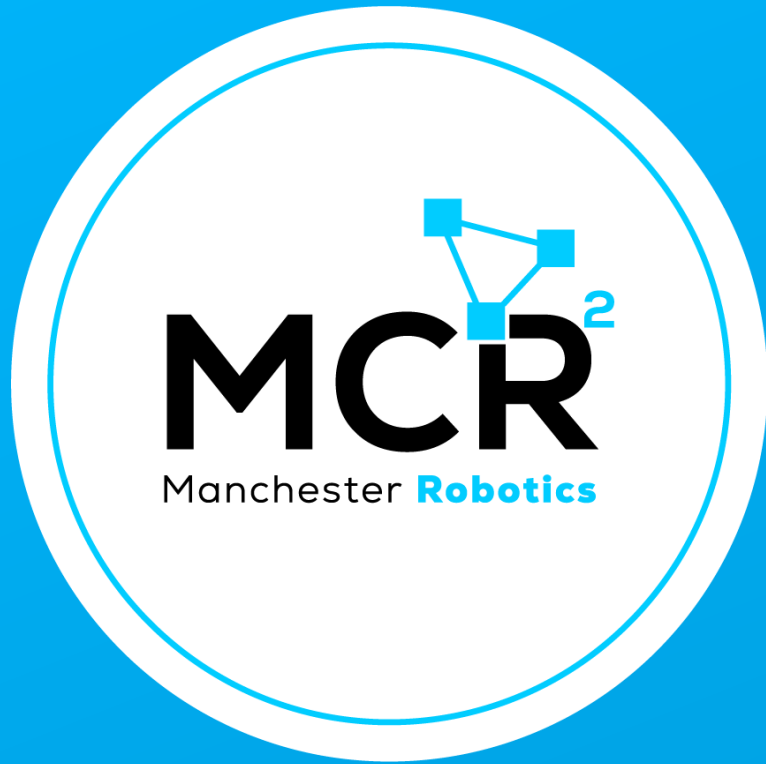
- Skills Gap
- Untrained employees
- Lack of practical training.

Hobbyists

- Too hardware-oriented.
- Lack of advanced applications and skills.
- The bigger picture is lost.

Higher Education, industry/corporates and Hobbyists are not keeping up with the rapid change and adoption in AI and Robotics

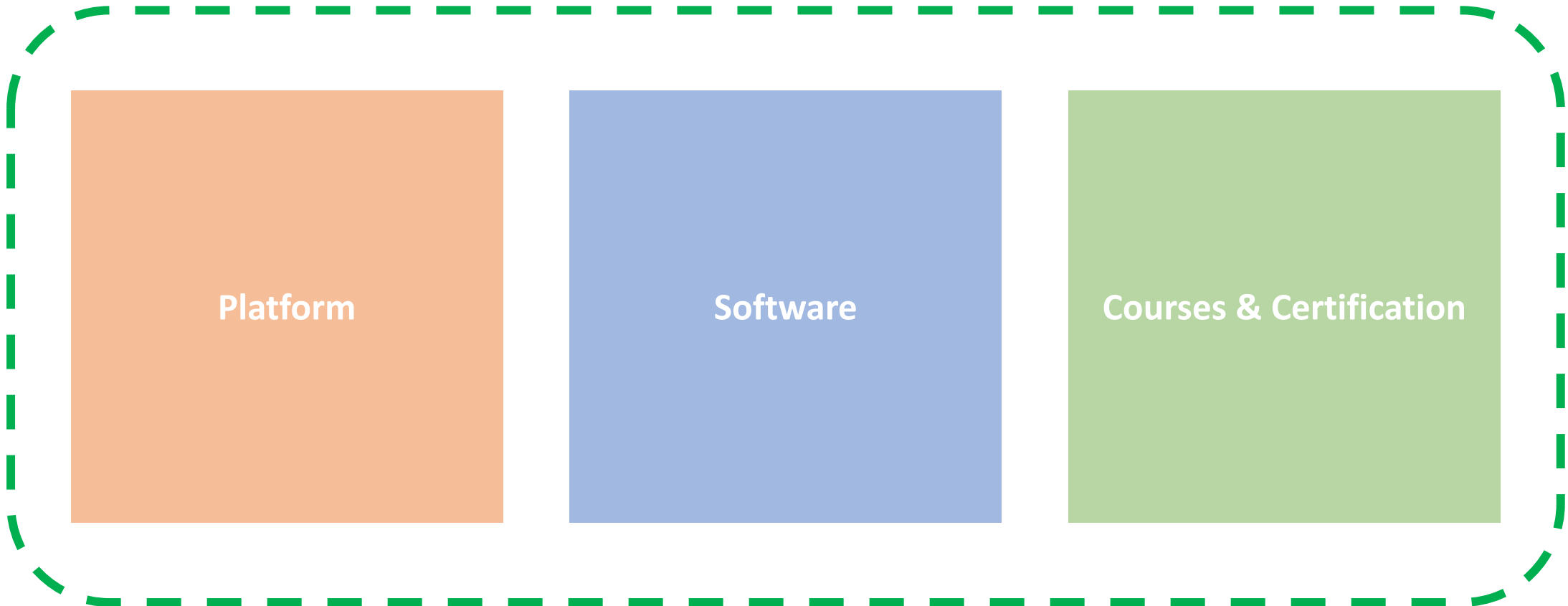




Our Solution



Our Solution: The Educational Robotics Ecosystem





Our Solution: The Educational Robotics Ecosystem

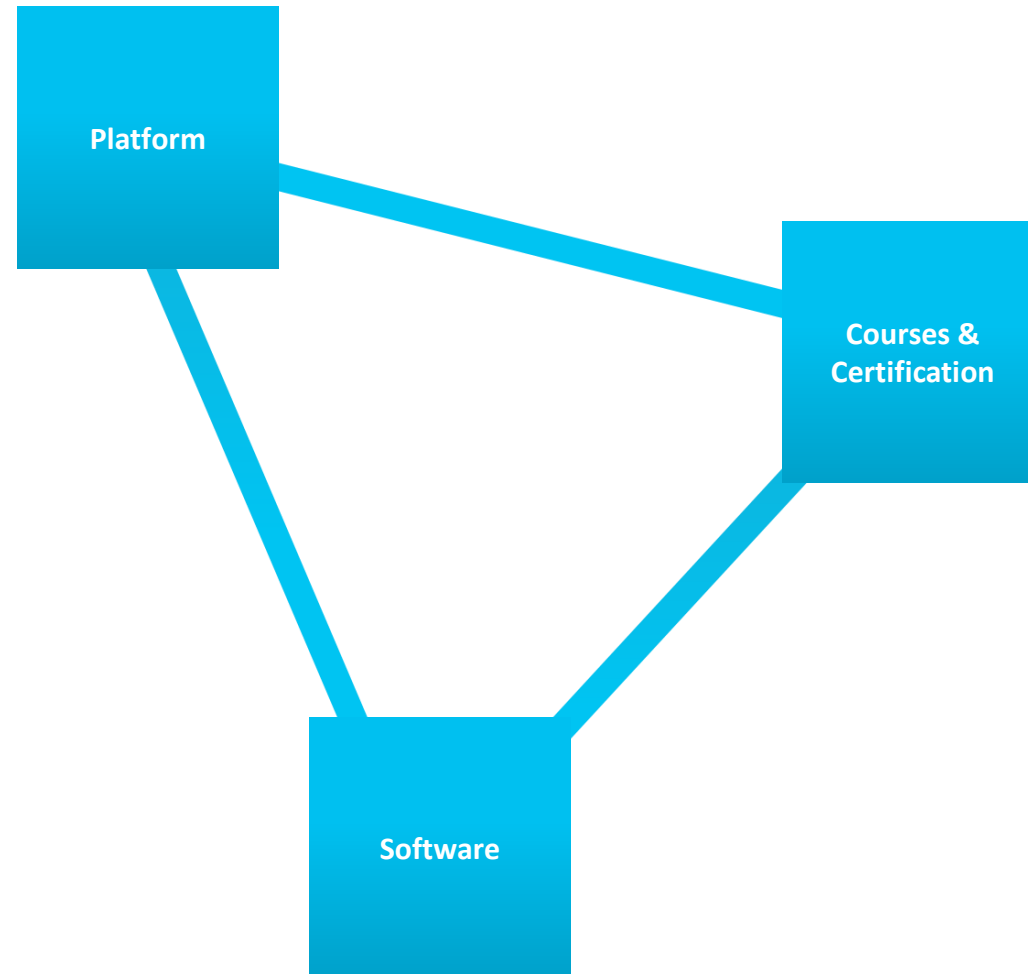
Platform

Software

Courses & Certification

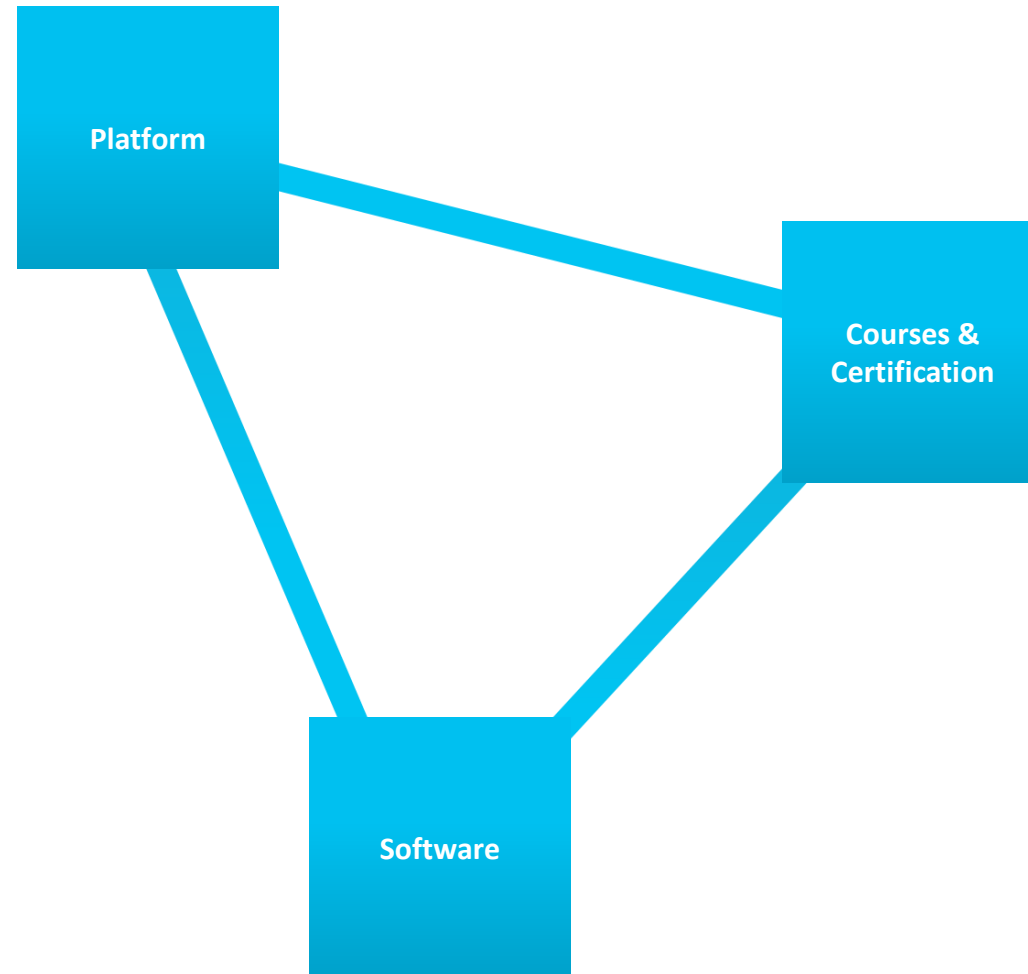


Our Solution: The Educational Robotics Ecosystem



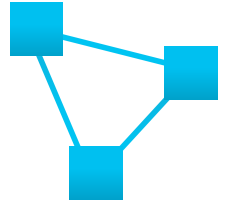


Our Solution: The Educational Robotics Ecosystem



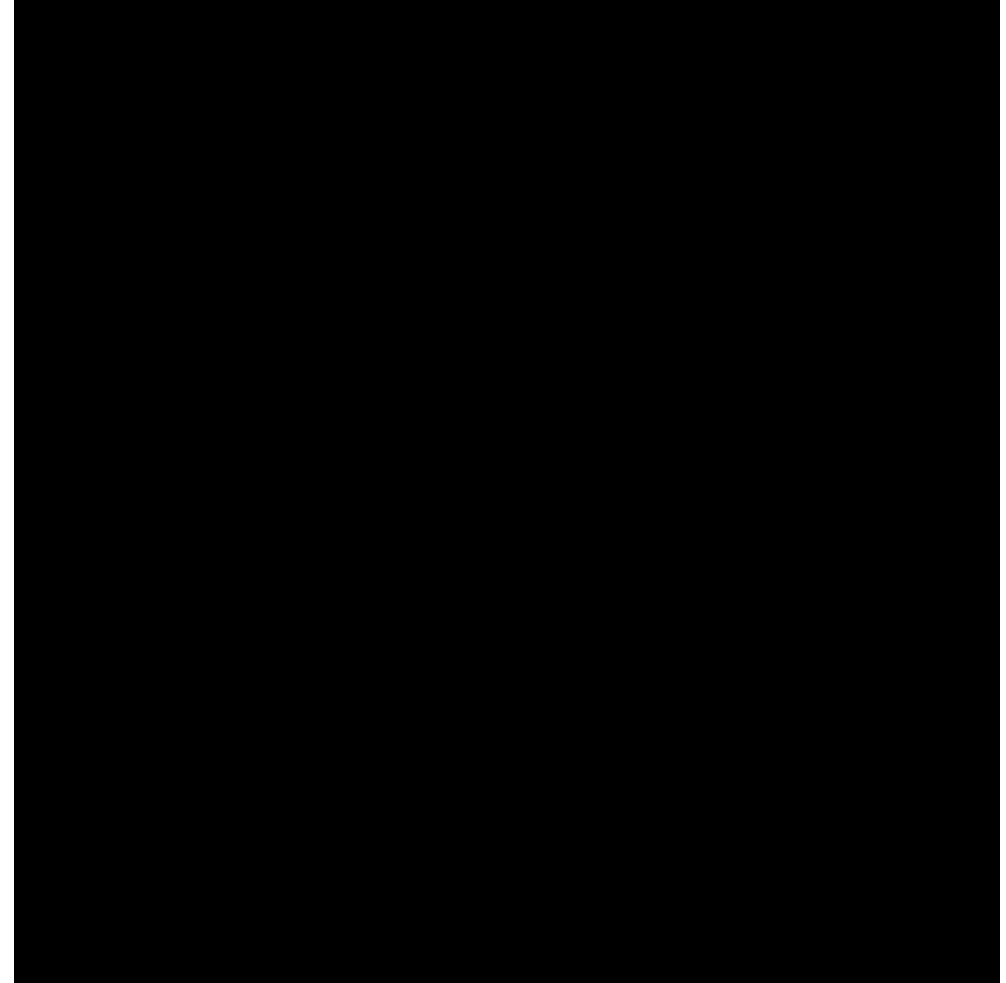


Our Solution: Platform



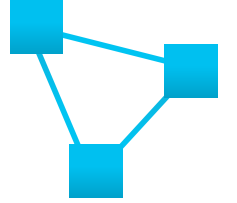
The Puzzlebot

- Affordable, customisable, modular, and open-source platform.
- Compatible with Python, MATLAB, LabVIEW, and ROS.
- Designed for beginners to advanced researchers.





Our Solution: Software



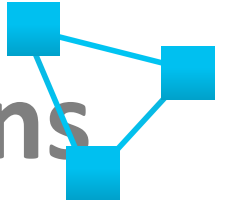
Puzzlebot SDK Suite

- A complete robotics SDK that takes you through the path of “[The Autonomy Hierarchy](#)”.
- Three-tier learning approach: Simulator, real-world testing, and industry application.
- Vertical integration: Start from zero knowledge and progress to AI-based robotics.
- Designed for students, researchers, and industry professionals.

Puzzlebot SDK



Our Solution: Courses & Certifications



Manchester Robotics Learning Path

Basic Certification: Introduction to robotics, programming, and control.

Professional Certification: Intermediate robotics applications, including AI and SLAM.

Specialist Certification: Advanced robotics, perception, and decision-making.

Ambassador Certification: Robotics leadership, research, and real-world applications.





What Makes Us Different?

Versatile Feature-set

Our circuit board and software are designed to be versatile to accommodate add-on components.

Advanced Capability

The circuit board is designed around powerful microprocessors and microcontrollers.

Accessible Price Point

We design with the intent of manufacturing at high volume to keep unit costs low.

Basic to Advanced Courses

Basic to advanced robotics courses developed alongside our partnership with NVIDIA

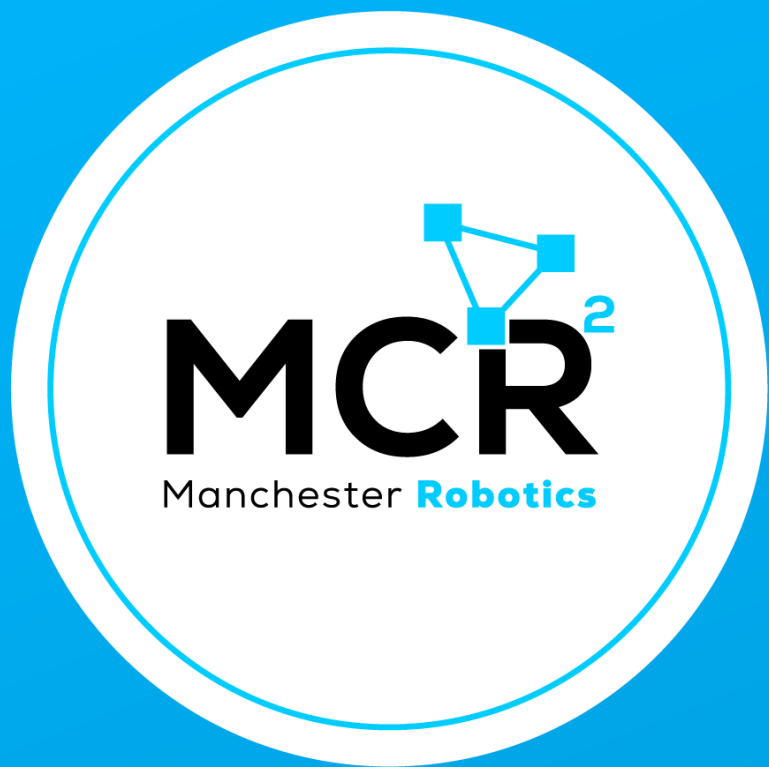
Solve real problems from industry

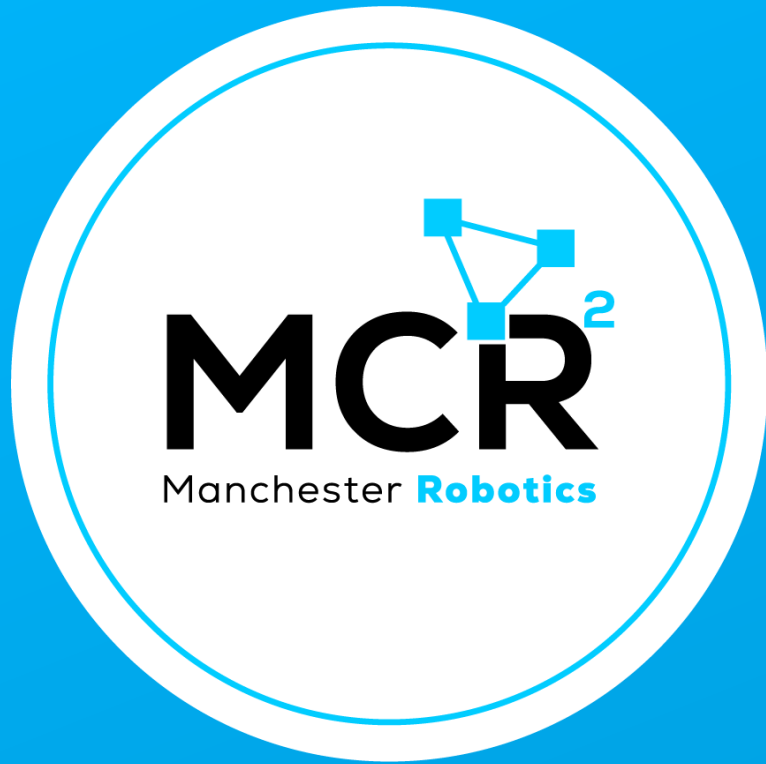
To help primary & secondary sectors of the global economy rapidly adopting robotics.

Challenge based approach

To provide challenges at each block, to engage the learners and improve their skills.







Real-World Impact

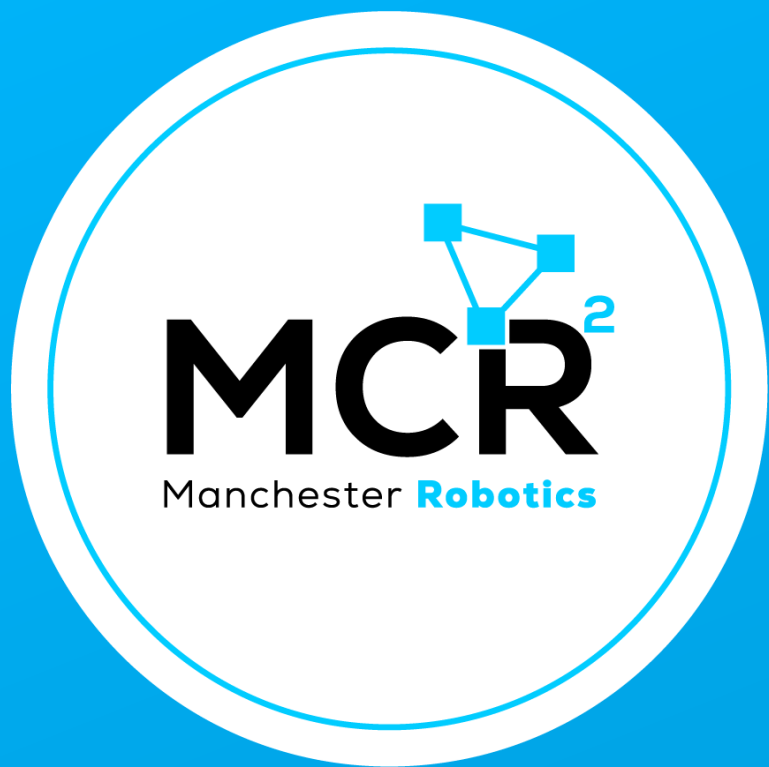


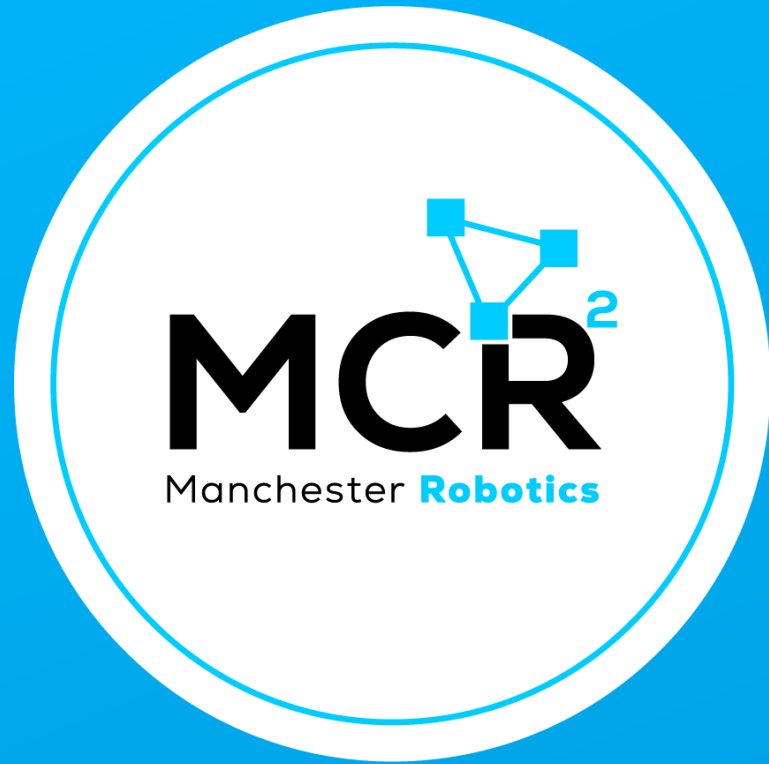
Real-World Impact

Global Reach & Industry Adoption

- University of Manchester, Durham University, and others.
- In Mexico: MCR2 Collaborates with UPY, Tec de Monterrey, Tecnológico Nacional de México.
- Collaboration with NVIDIA to integrate AI and robotics education.
- Published papers, research projects, and industry partnerships.







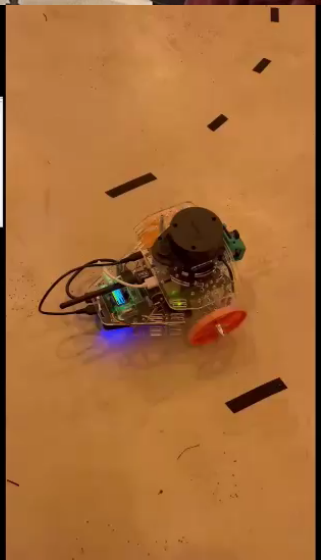
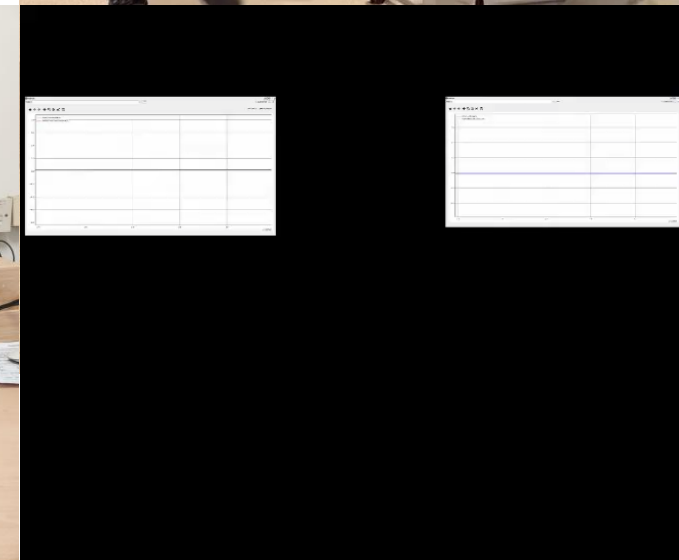
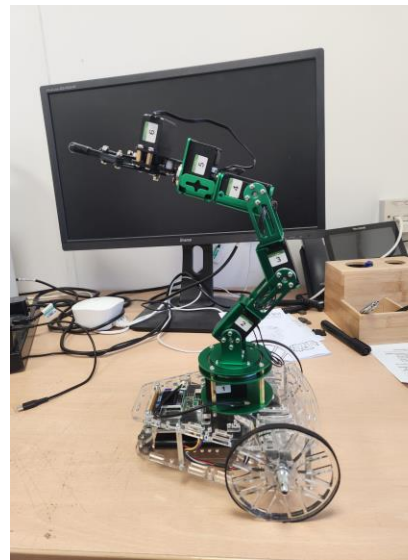
New Developments



New Developments

Expanding the Ecosystem

- Puzzlebot Drone Ed- An educational drone for aerial robotics.
- Puzzlebot Arm - Extending capabilities for manipulation tasks.
- Interactive Robotics Experience - A fully immersive way to learn robotics through digital twins and real-world testing.





Partnerships: Tec de Monterrey

- Manchester Robotics has established partnerships with different educational institutions worldwide.
- In Mexico, we've partnered with Tecnológico de Monterrey for more than 4 years now.
- Different project and courses have been done in different campuses nationwide alongside professors and students from different backgrounds.





Partnerships: Tec de Monterrey

- Papers from education and robotics innovation, have been published in different journals.
- Research is currently being done using MCR2 Tools in the fields of Mechatronics and Robotics.

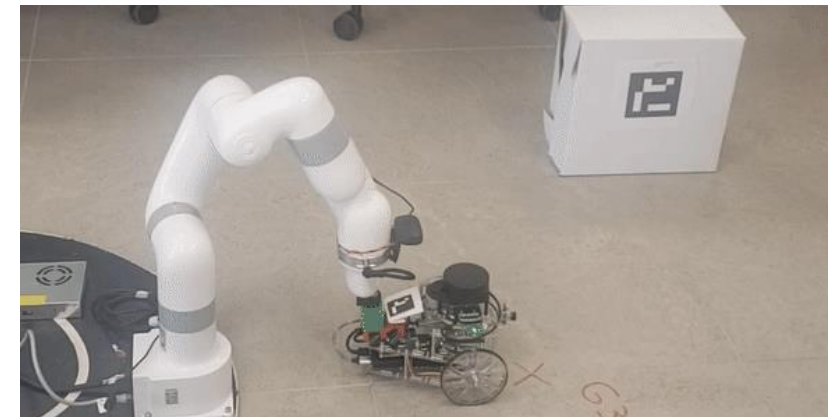
A study on teaching Cyber-Physical Systems with a customized branded mobile robot for Industry 4.0

Consuelo Rodríguez-Padilla¹, Mario Martínez Guerrero², Alexandru Stancu³, Karla Yokoyani Chavero Valencia⁴, Bernardo Flores Reyes¹,
Jerem y Bruce Taylor Valdez¹, Carlos Vazquez Hurtado¹
¹Tecnológico de Monterrey, Escuela de Ingeniería y Ciencias, Ave. Eugenio Garza Sada 2501, Monterrey 64849, NL, Mexico
[corodriguez, A01659228, A01029744, A01751664, cvazquezh] @tec.mx
²CTO / Co-Founder of Manchester Robotics Ltd., University of Manchester, UK, mario@manchester-robotics.com
³Department of Electrical and Electronic Engineering and the Robotics and Autonomous Systems Group Leader at the Aerospace Research
Institute, University of Manchester, UK, alexandru.stancu@manchester.ac.uk

**3D Gaussian Splatting for Emulating
Real Environments in Undergraduate
Educational Robotics.**

Arturo E. Cerón, Jason A. Castaño

 **Tecnológico
de Monterrey**
June 2024



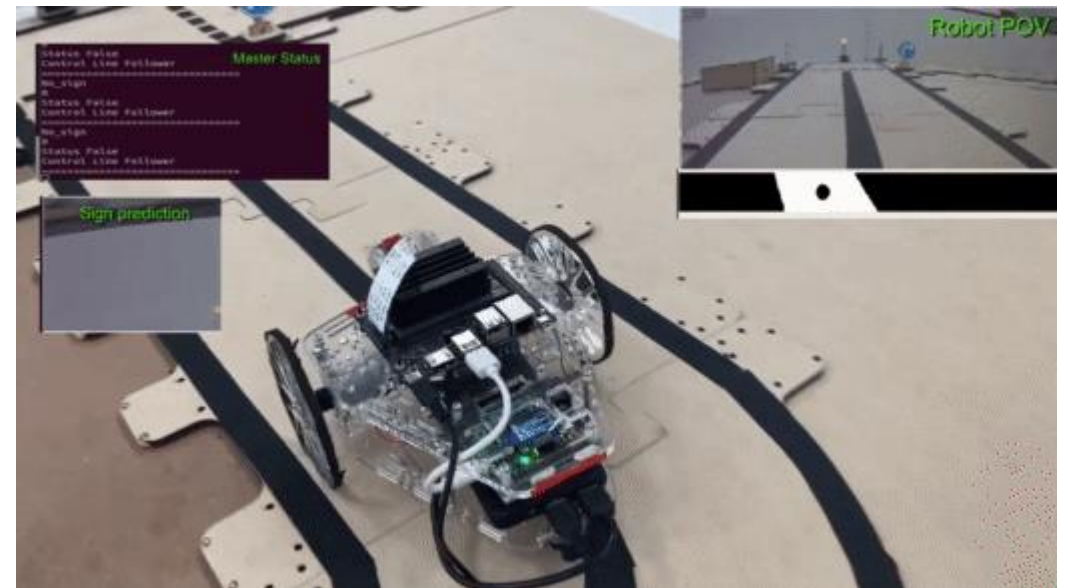
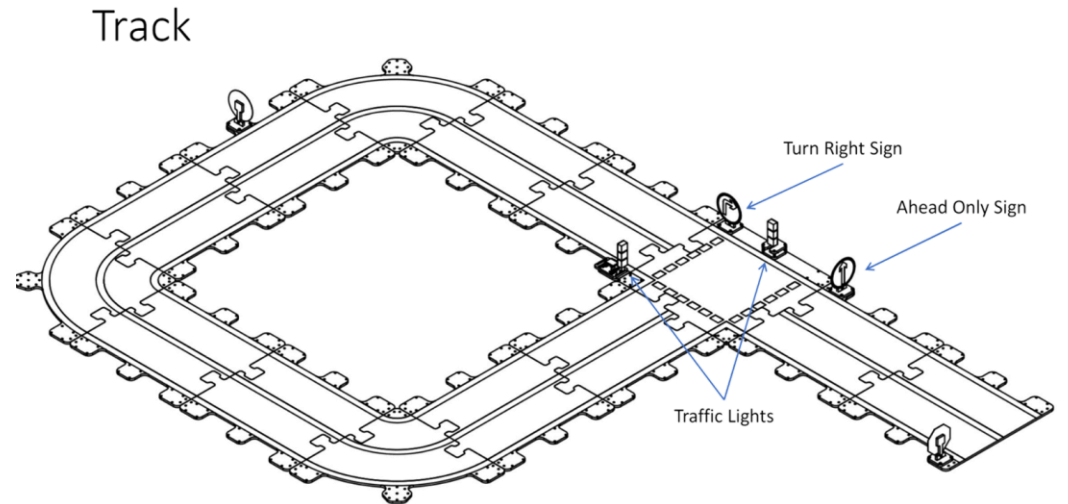


Partnerships: Tec de Monterrey



Tec de Monterrey: Nationwide Robotics Challenge

- Consists in autonomously drive the robot on a predefined track provided by MCR2.
- Using the knowledge acquired during the semester, the students were able to implement a neural networks alongside some image recognition algorithms and control theory to drive the robot on the track while obeying the traffic signals.
- The best teams got the chance exhibit their results to their professors, the directors of MCR2, and the engineering directors at NVIDIA.

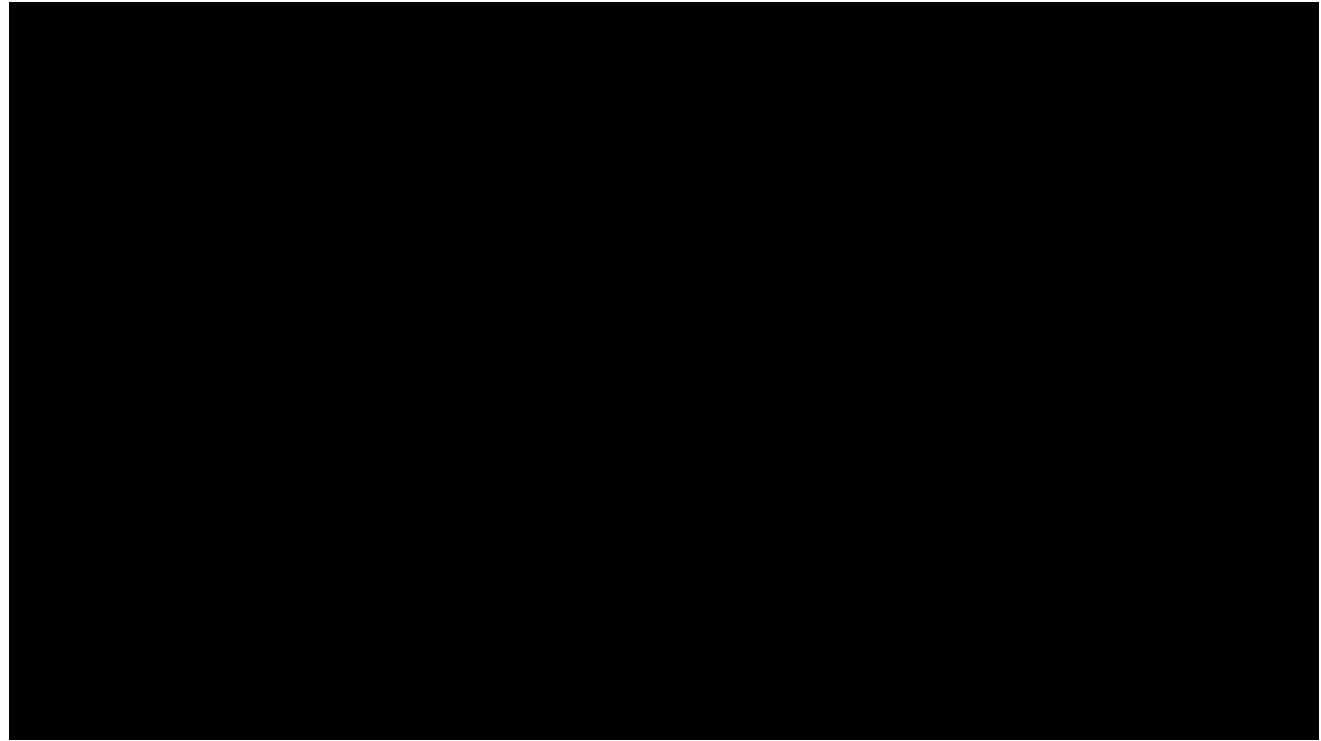


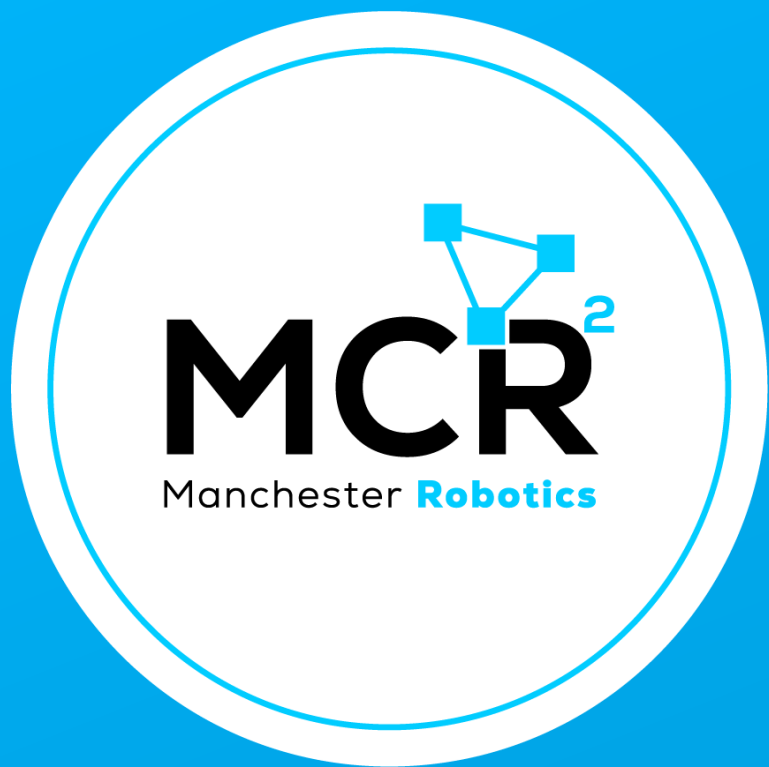


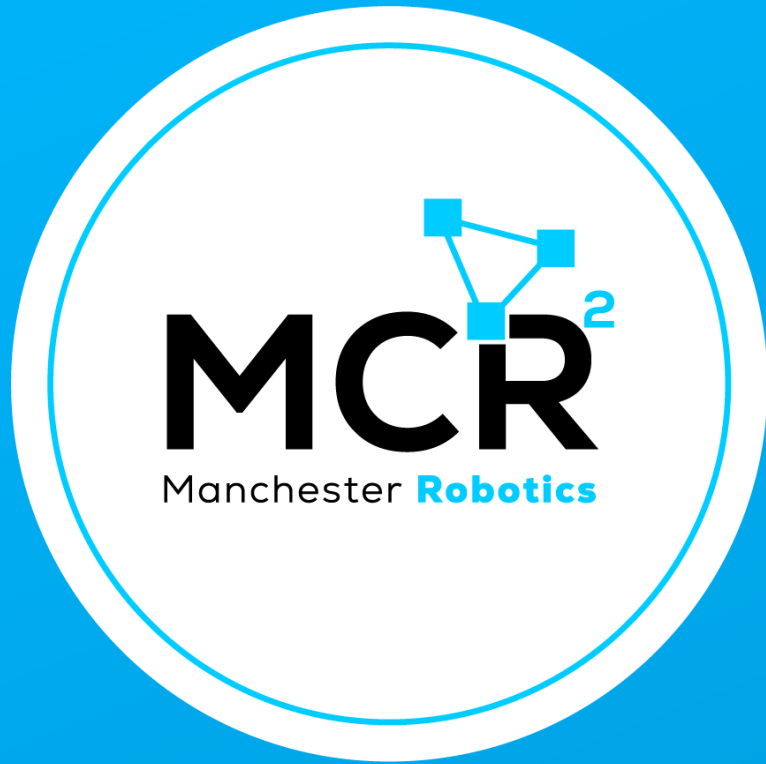
Partnerships: Tec de Monterrey

Tec de Monterrey Automation Challenge

- Making a production line in accordance with MCR2 standards.
- Using the knowledge acquired during the semester, the students were able to implement some image recognition algorithms, manufacturing and control theory to implement the manufacturing line.





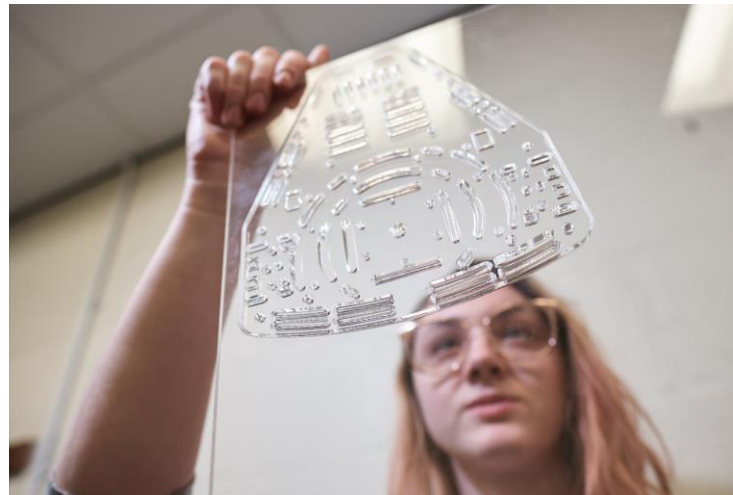
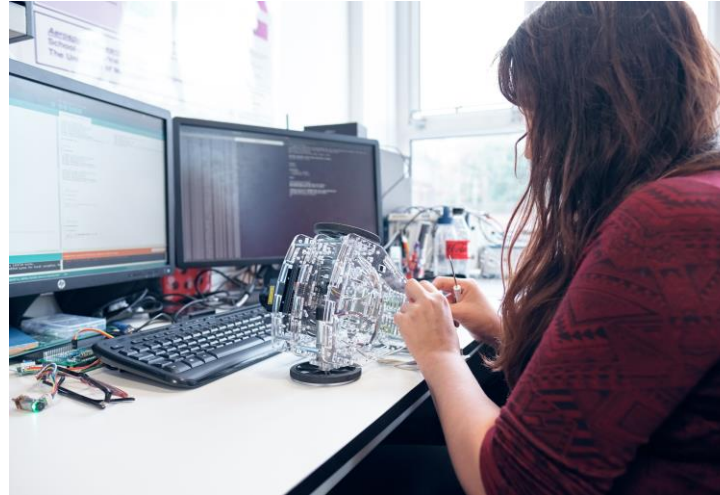


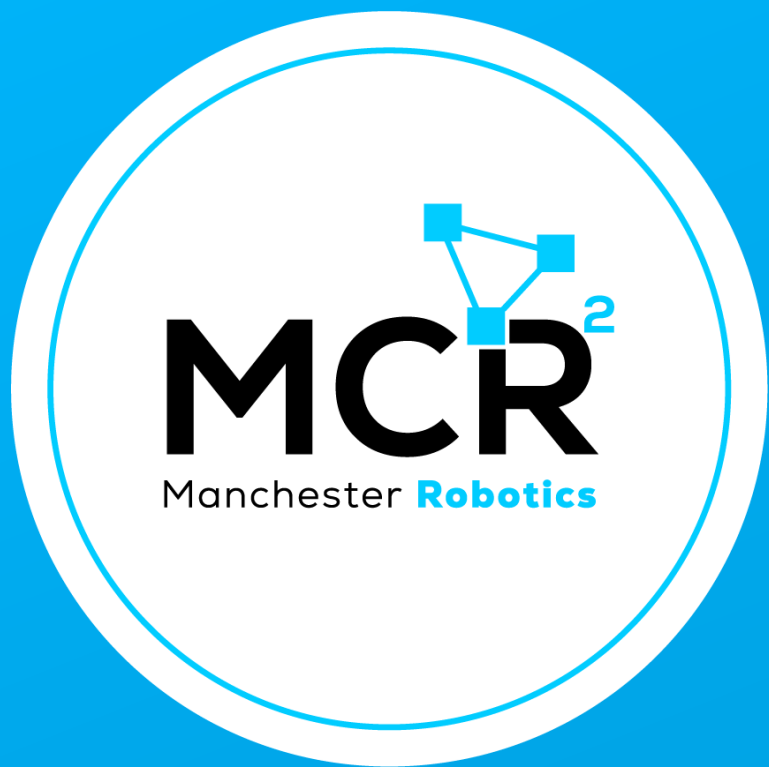
Future Vision

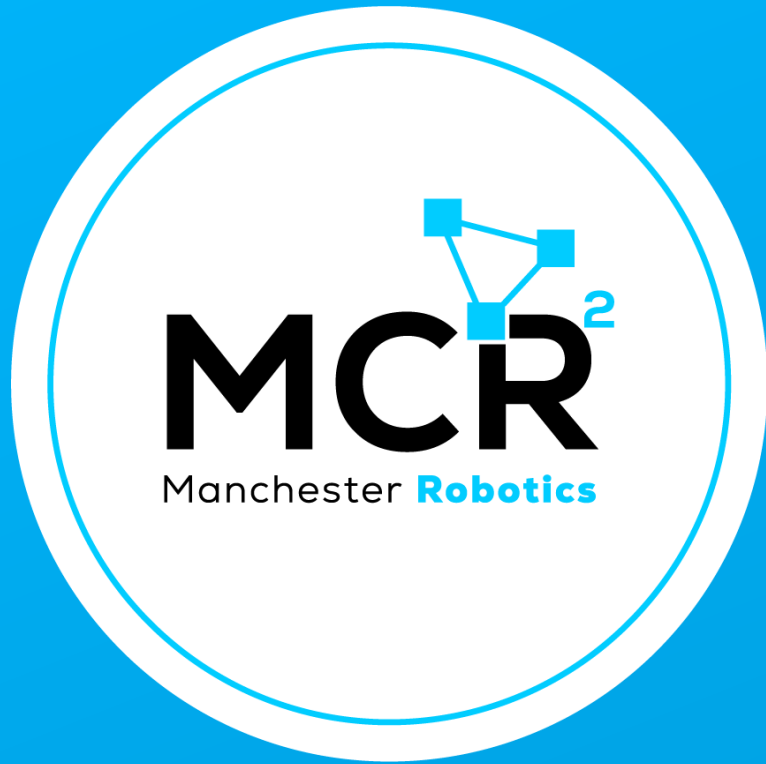


Future Vision

- Expanding to more universities and industries.
- Making Puzzlebot Ecosystem a global standards in education.
- Creating an AI-powered robotics learning assistant.







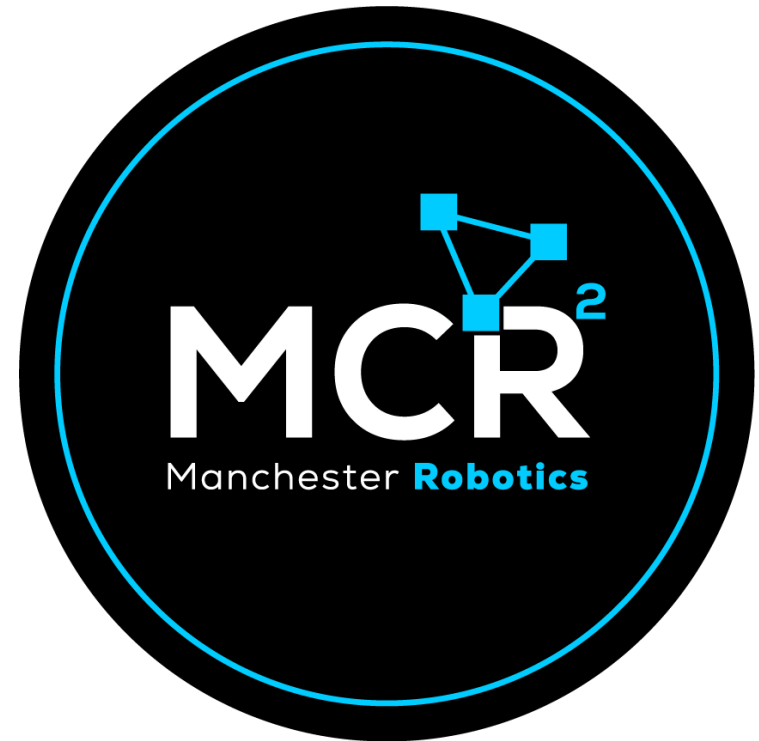
Call to Action

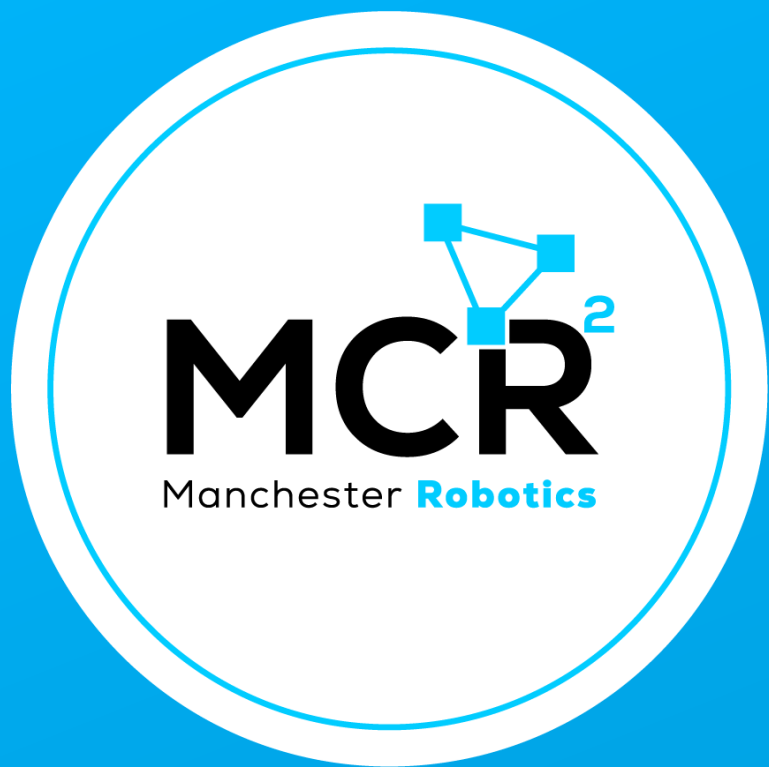


Call to Action

Join the Revolution!

- Students & Educators: Contact us to get involved with Puzzlebot and our robotics programs.
- Researchers & Industry Partners: Collaborate with us to develop new applications.
- Investors & Entrepreneurs: Support the mission to democratize robotics education.

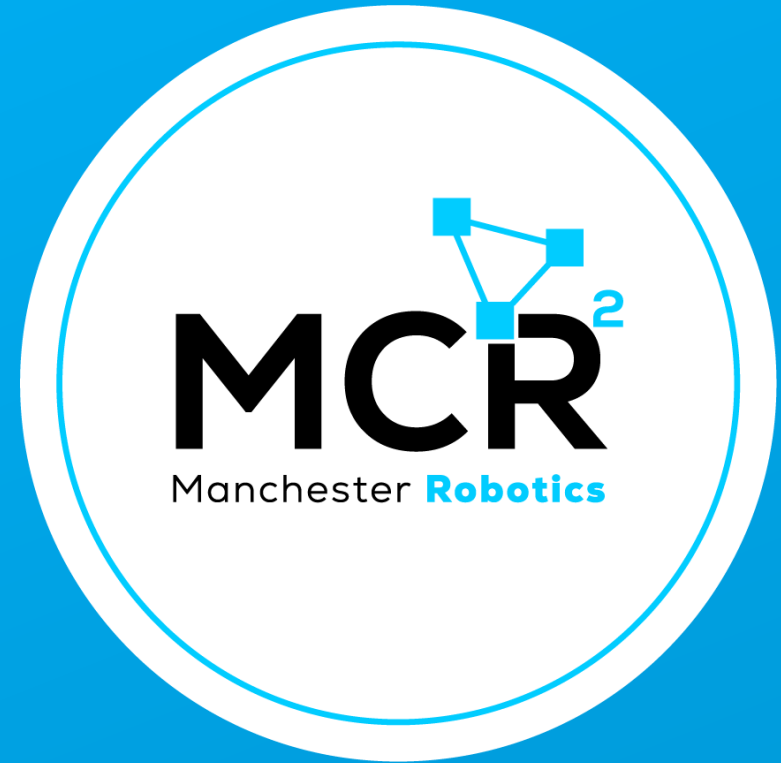






Thank You

Robotics For Everyone



{Learn, Create, Innovate};